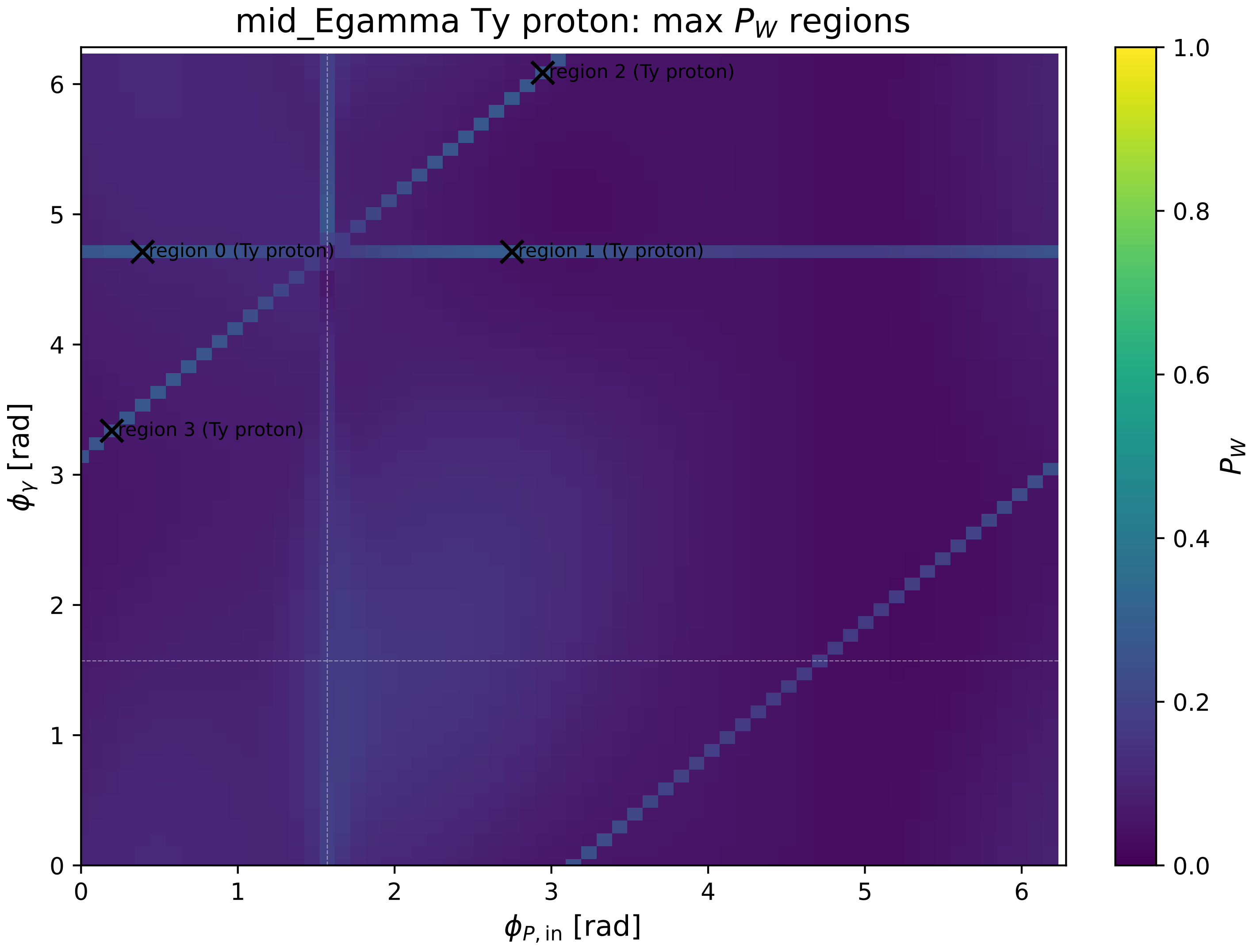
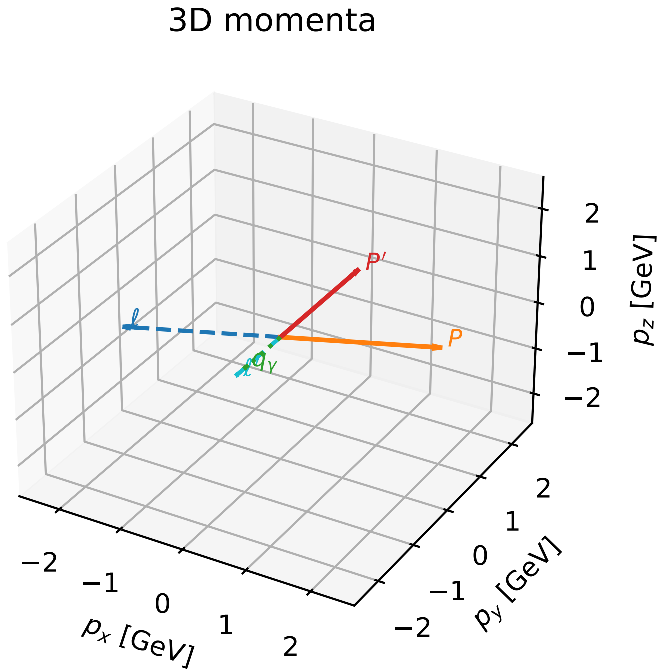

mid_Egamma Ty proton: max P_W regions
region 2 (Ty proton)
region 0 (Ty proton)
region 1 (Ty proton)
region 3 (Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

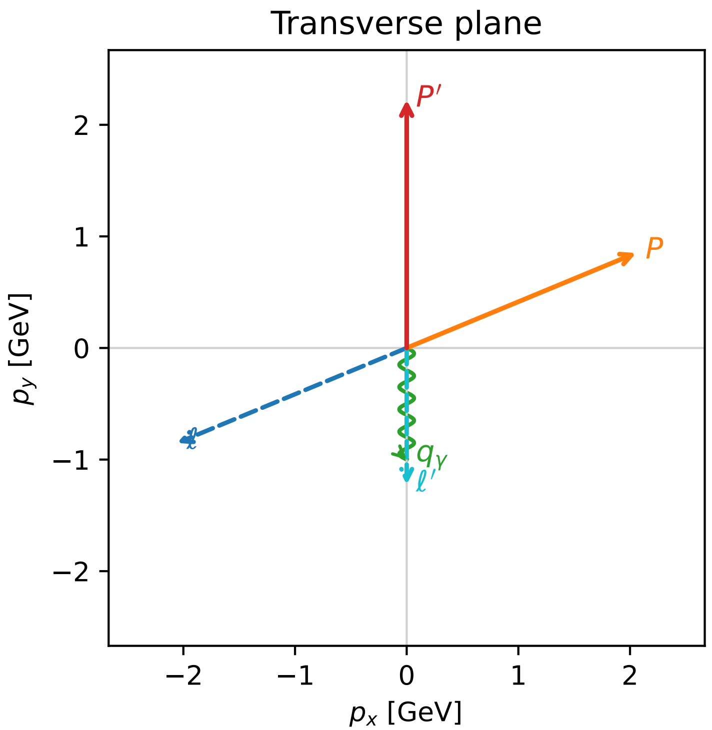
Ty proton characteristic max P_W configuration

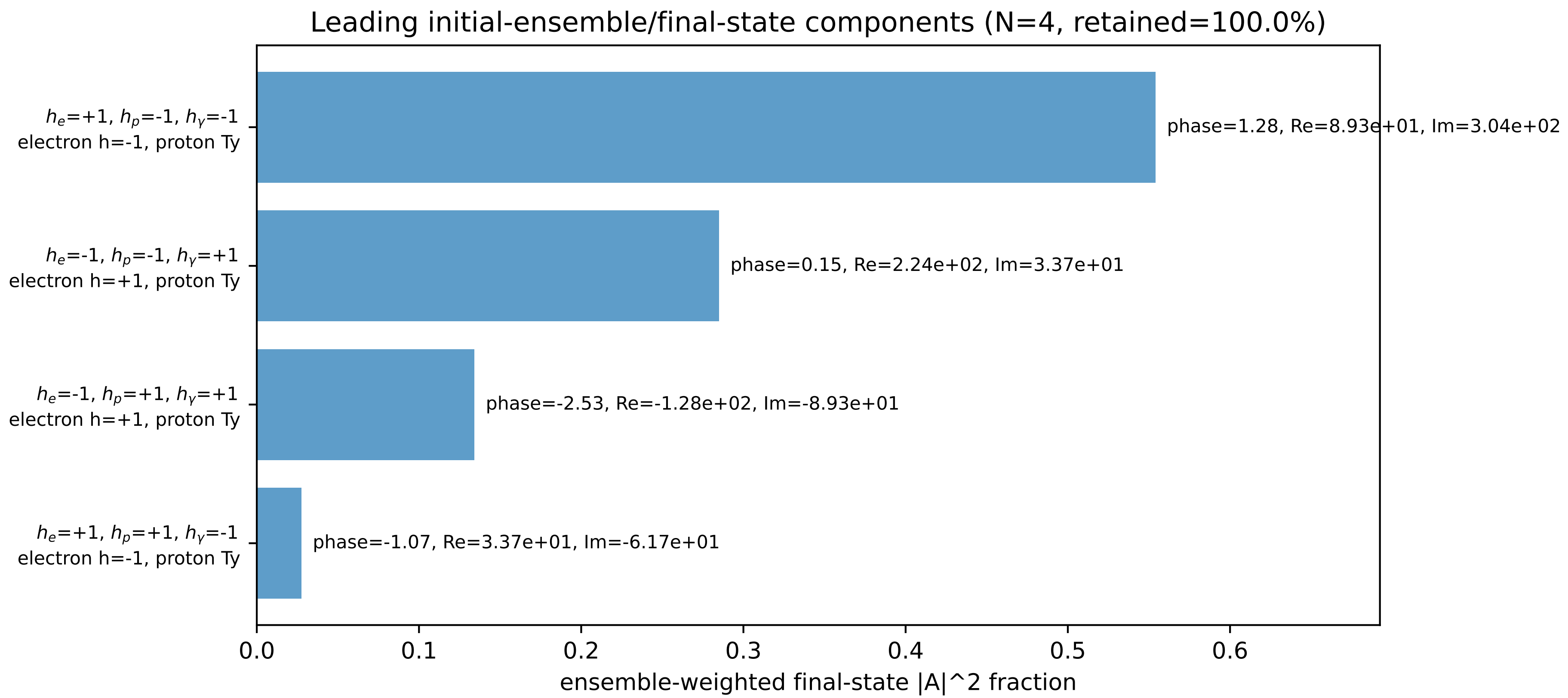


w_purity_mid_egamma_ty_proton_region_0 P_W=0.279659
region: mid_Egamma_Ty_proton_region_0
outgoing-state purity: 0.513201
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
θ_{in}=1.57079, ϕ_ℓ=3.53429, ϕ_P=0.392699
E_γ=1, ϕ_γ=4.71239
Q^2=3.36267, x_B=0.26604, t=-6.10901, W^2=10.1569, y=0.612507
$\theta(\ell', \gamma)$=0 deg

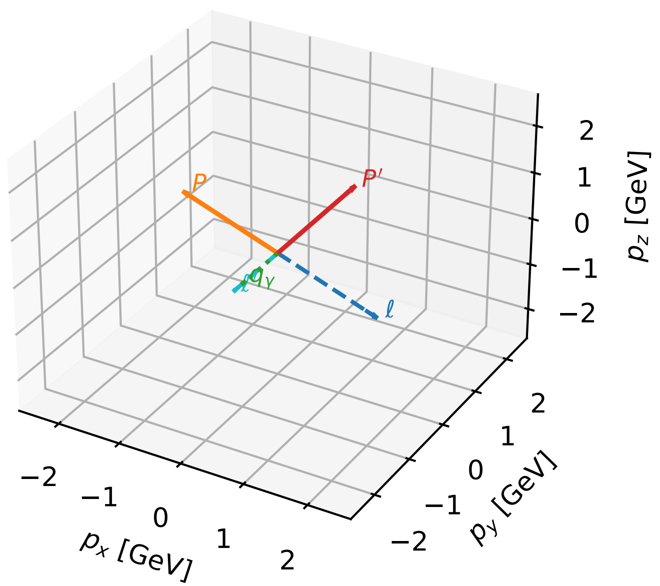
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -2.0551 p_y= -0.8512 p_z= -0.0000
P E= 2.4141 p_x= 2.0551 p_y= 0.8512 p_z= 0.0000
ℓ' E= 1.2244 p_x= 0.0000 p_y= -1.2244 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000





Ty proton characteristic max P_W configuration

3D momenta

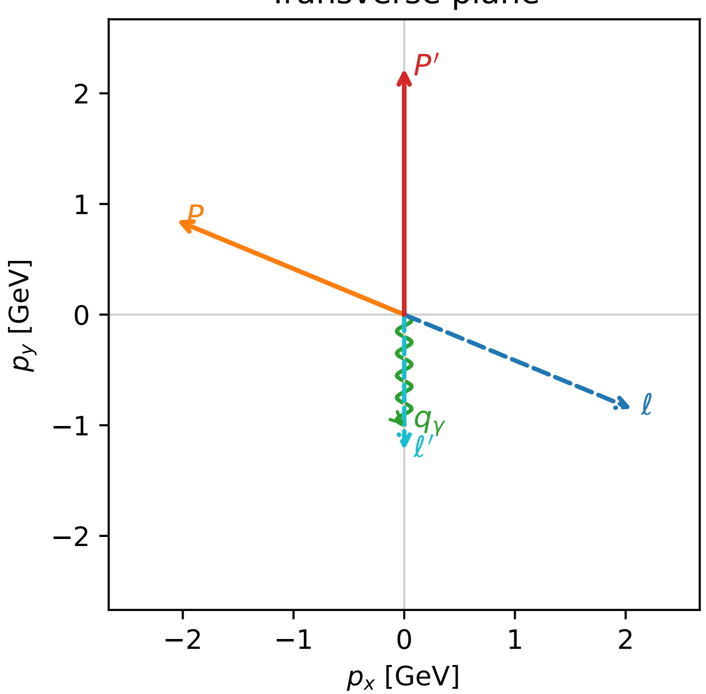


w_purity_mid_egamma_ty_proton_region_1 P_W=0.279522
region: mid_Egamma_Ty_proton_region_1
outgoing-state purity: 0.513201
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$

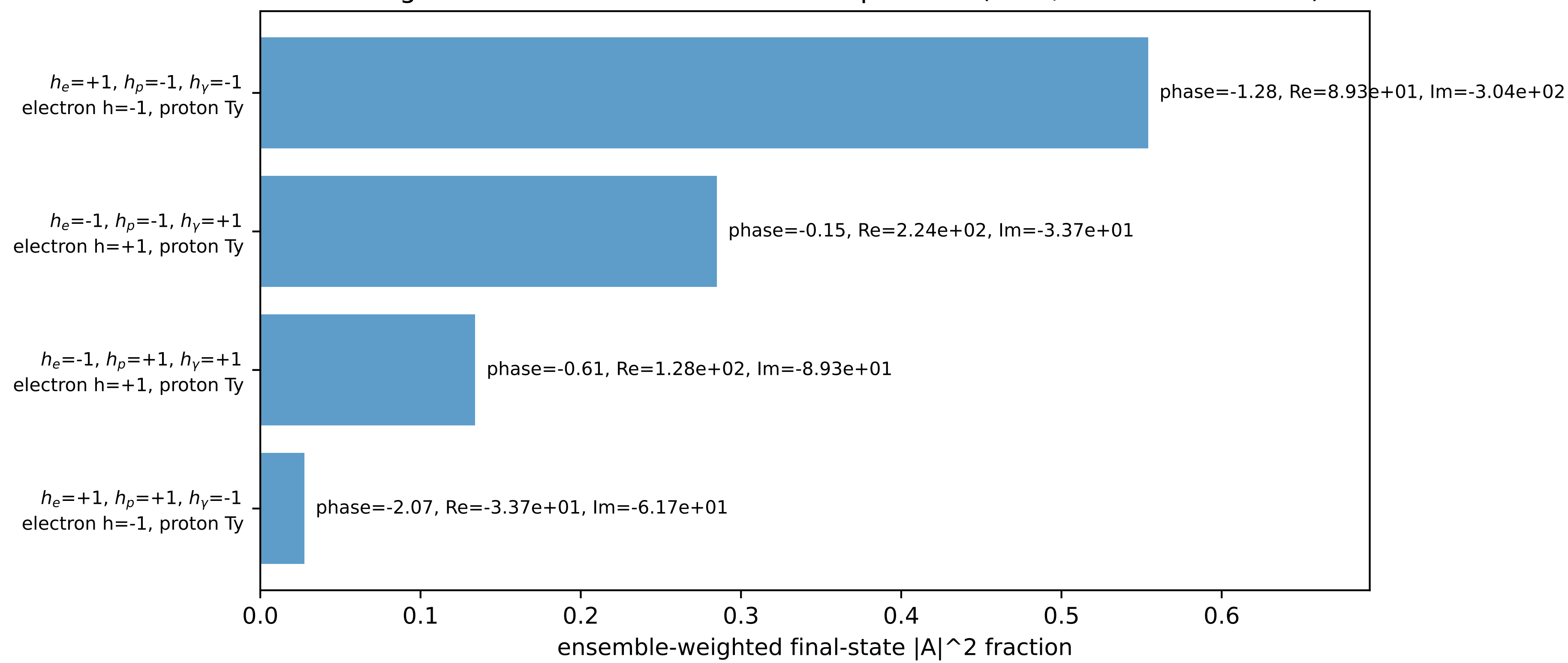
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
$\theta_{\rm in}$=1.57079, ϕ_ℓ=5.89049, ϕ_P=2.74889
E_γ=1, ϕ_γ=4.71239
Q^2=3.36267, x_B=0.26604, t=-6.10901, W^2=10.1569, y=0.612507
$\theta(\ell', \gamma)$=0 deg

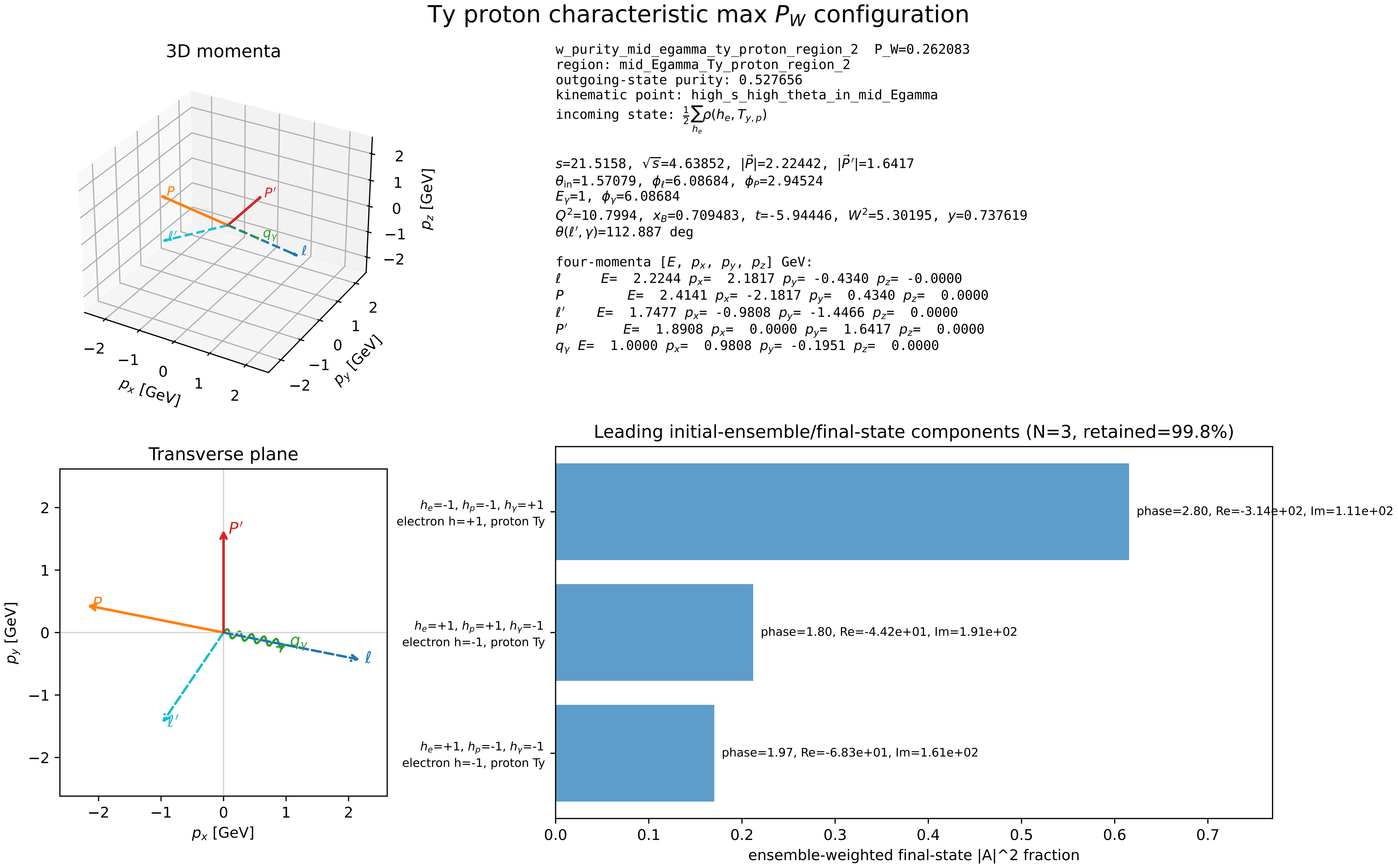
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.0551 p_y= -0.8512 p_z= -0.0000
P E= 2.4141 p_x= -2.0551 p_y= 0.8512 p_z= 0.0000
ℓ' E= 1.2244 p_x= 0.0000 p_y= -1.2244 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)





Ty proton characteristic max P_W configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
w_purity_mid_egamma_ty_proton_region_2 P_W=0.262083
region: mid_Egamma_Ty_proton_region_2
outgoing-state purity: 0.527656
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.6417
$\theta_{\rm in}$=1.57079, ϕ_ℓ=6.08684, ϕ_P=2.94524
E_γ=1, ϕ_γ=6.08684
Q^2=10.7994, x_B=0.709483, t=-5.94446, W^2=5.30195, y=0.737619
$\theta(\ell',\gamma)$=112.887 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.1817 p_y= -0.4340 p_z= -0.0000
P E= 2.4141 p_x= -2.1817 p_y= 0.4340 p_z= 0.0000
ℓ' E= 1.7477 p_x= -0.9808 p_y= -1.4466 p_z= 0.0000
P' E= 1.8908 p_x= 0.0000 p_y= 1.6417 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.9808 p_y= -0.1951 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=3, retained=99.8%)
h_e=-1, h_p=-1, h_γ=+1
electron h=+1, proton Ty
phase=2.80, Re=-3.14e+02, Im=1.11e+02
h_e=+1, h_p=+1, h_γ=-1
electron h=-1, proton Ty
phase=1.80, Re=-4.42e+01, Im=1.91e+02
h_e=+1, h_p=-1, h_γ=-1
electron h=-1, proton Ty
phase=1.97, Re=-6.83e+01, Im=1.61e+02
ensemble-weighted final-state |A|^2 fraction

Ty proton characteristic max P_W configuration